

ATSA Unit 3 Q&A Notes

Stationary Process and ARMA Models | Simple exam-focused notes

Use these notes for quick learning, voice revision, and PYQ practice. The answers are written in simple exam language based on the Unit 3 PDF topics.

How To Study This Unit

- First understand stationarity and white noise.
- Then learn differencing and backshift notation.
- After that learn AR, MA, ARMA, ACF and PACF.
- Finally practise PYQ answers and autocorrelation numericals.

Unit 3 Topic Checklist

No.	Topic	Priority
1	Random walk model	High
2	Stationarity	Very High
3	Non-stationarity in the mean	High
4	Differencing	Very High
5	Second-order differencing	Medium
6	Seasonal differencing	High
7	First order difference plus seasonal difference	High
8	Backshift notation	Very High
9	Autoregressive models, AR(1), AR(2)	Very High
10	Moving Average models, MA(1), MA(2), MA(infinity)	Very High
11	Invertibility	Medium
12	Partial autocorrelations	Very High
13	ACF and PACF interpretation	Very High
14	White noise	Very High

Q&A Notes

Q. What is a random walk model?

A random walk is a time series where the current value depends on the previous value plus a random error.

Model: $y_t = y_{(t-1)} + e_t$

Meaning: The next value moves from the last value by a random shock e_t .

Important point: A random walk is usually non-stationary because shocks keep accumulating over time.

Exam line: Random walk is a common example of a non-stationary time series.

Q. What is stationarity?

Stationarity means the statistical behavior of the time series remains stable over time.

Weak stationarity conditions: constant mean, constant variance, and autocovariance depends only on lag.

Values can go up and down, but the overall mean and spread should not keep changing with time.

Exam line: ARMA models are used for stationary time series.

Q. What is white noise?

White noise is a purely random time series with no predictable pattern.

Conditions: mean is zero or constant, variance is constant, and autocorrelation is zero at all non-zero lags.

In modelling, residuals should behave like white noise because it means the model captured the useful pattern.

Exam line: If residuals are white noise, the fitted model is usually considered adequate.

Q. What is non-stationarity in the mean?

Non-stationarity in the mean means the average level of the series changes over time.

Example: sales increasing every month due to trend. The mean is not constant.

This can often be handled by differencing or detrending.

Exam line: If a series has trend, mention that it is non-stationary in mean.

Q. What is differencing?

Differencing means subtracting the previous value from the current value.

Formula: First difference: $\Delta y_t = y_t - y_{(t-1)}$

It converts a level series into a change series and helps remove trend.

Exam line: Differencing is used to make a non-stationary series stationary.

Q. How does first-order differencing remove a linear trend?

Given: $y_t = \beta_0 + \beta_1 t + e_t$

First difference: $\Delta y_t = y_t - y_{(t-1)}$

Substitution: $\Delta y_t = [\beta_0 + \beta_1 t + e_t] - [\beta_0 + \beta_1(t-1) + e_{(t-1)}]$

Result: $\Delta y_t = \beta_1 + e_t - e_{(t-1)}$

The time trend t is removed. So first differencing can make a trend-stationary series stationary.

Exam line: This exact proof appeared in the current paper style.

Q. What is second-order differencing?

Second-order differencing means differencing the already differenced series once again.

Formula: $\Delta^2 y_t = \Delta y_t - \Delta y_{(t-1)}$

Expanded form: $\Delta^2 y_t = y_t - 2y_{(t-1)} + y_{(t-2)}$

It is used when first differencing is not enough to remove non-stationarity.

Exam line: Do not over-difference; it can make the model unnecessarily complex.

Q. What is seasonal differencing?

Seasonal differencing subtracts the value from the same season in the previous cycle.

Formula: $y_t - y_{(t-s)}$

Example: For monthly data with yearly seasonality, $s = 12$. For quarterly data, $s = 4$.

It removes repeating seasonal patterns.

Exam line: Use seasonal differencing when the series has seasonal non-stationarity.

Q. What is first-order difference plus seasonal difference?

Sometimes the data has both trend and seasonality.

First-order differencing removes trend, and seasonal differencing removes seasonal pattern.

Combined form: $(1 - B)(1 - B^s)y_t$

Exam line: This is used before fitting seasonal ARIMA models.

Q. What is backshift notation?

Backshift operator B shifts the series one time step backward.

Rules: $B y_t = y_{(t-1)}$, $B^2 y_t = y_{(t-2)}$, and $B^s y_t = y_{(t-s)}$.

Differencing: $(1 - B)y_t = y_t - y_{(t-1)}$.

Backshift notation makes ARMA and ARIMA equations shorter.

Exam line: Backshift notation is very important for ARIMA/SARIMA equation questions.

Q. What is an autoregressive model?

An autoregressive model predicts the current value using previous values of the same series.

General AR(p) model: $y_t = c + \phi_1 y_{(t-1)} + \phi_2 y_{(t-2)} + \dots + \phi_p y_{(t-p)} + e_t$

Meaning of p: p is the number of past values used.

Exam line: AR means current value depends on past actual values.

Q. What is AR(1)?

AR(1) is an autoregressive model of order 1.

Model: $y_t = c + \phi_1 y_{(t-1)} + e_t$

It uses only one previous value to predict the current value.

Exam line: In AR(1), PACF cuts off after lag 1 and ACF tails off.

Q. What is the AR(1) model condition for stationarity?

For AR(1), the model is stationary if the coefficient is between -1 and +1.

Condition: $|\phi_1| < 1$

If the coefficient is too large, the effect of past values does not die out properly.

Exam line: Write $|\phi_1| < 1$ for AR(1) stationarity.

Q. What is AR(2)?

AR(2) uses two previous values to predict the current value.

Model: $y_t = c + \phi_1 y_{(t-1)} + \phi_2 y_{(t-2)} + e_t$

It is useful when the current value depends on the previous two time points.

Exam line: In AR(2), PACF usually cuts off after lag 2.

Q. What is a Moving Average model?

A Moving Average model predicts the current value using current and past error terms.

General MA(q) model: $y_t = \mu + e_t + \theta_1 e_{(t-1)} + \dots + \theta_q e_{(t-q)}$

Meaning of q: q is the number of past errors used.

Exam line: MA means current value depends on past forecast errors.

Q. What is MA(1)?

MA(1) is a moving average model of order 1.

Model: $y_t = \mu + e_t + \theta_1 e_{(t-1)}$

It uses the current error and one past error.

Exam line: In MA(1), ACF cuts off after lag 1 and PACF tails off.

Q. What is MA(2)?

MA(2) is a moving average model of order 2.

Model: $y_t = \mu + e_t + \theta_1 e_{(t-1)} + \theta_2 e_{(t-2)}$

It uses the current error and two past errors.

Exam line: In MA(2), ACF cuts off after lag 2.

Q. What is MA(infinity)?

MA(infinity) means the model uses infinitely many past error terms.

In theory, some AR models can be represented as MA(infinity) under suitable conditions.

It shows that AR and MA ideas are connected mathematically.

Exam line: Usually this is conceptual; do not spend too much time on it for the exam.

Q. What is invertibility?

Invertibility means an MA model can be rewritten as an AR(infinity) model in a stable and unique way.

It ensures that past shocks can be recovered from the observed series.

Invertibility is important for meaningful MA model estimation and forecasting.

Exam line: Write that invertibility makes the MA representation stable and forecastable.

Q. What is partial autocorrelation?

Partial autocorrelation measures the direct relationship between y_t and $y_{(t-k)}$ after removing the effect of shorter lags.

Example: PACF at lag 2 measures direct relation between y_t and $y_{(t-2)}$ after removing the effect of $y_{(t-1)}$.

PACF is mainly used to identify AR order.

Exam line: PACF cut-off helps select p in $AR(p)$.

Q. What are ACF and PACF?

ACF: Autocorrelation Function measures correlation between observations at different lags.

PACF: Partial Autocorrelation Function measures direct correlation at a lag after removing shorter lag effects.

Use: They help identify AR, MA, ARMA, and ARIMA orders.

Exam line: ACF/PACF appeared in both papers.

ACF/PACF Identification Rules

$AR(p)$: PACF cuts off after lag p ; ACF tails off gradually.

$MA(q)$: ACF cuts off after lag q ; PACF tails off gradually.

$ARMA(p,q)$: both ACF and PACF tail off.

Memory trick: PACF cut = AR. ACF cut = MA.

Important Formula Sheet

Core Unit 3 Formulas

Mean: $\bar{x} = \sum(x_t) / n$

Variance: $\gamma_0 = \sum[(x_t - \bar{x})^2] / n$

Autocovariance at lag k : $\gamma_k = \sum[(x_t - \bar{x})(x_{(t-k)} - \bar{x})] / n$

Autocorrelation at lag k : $r_k = \gamma_k / \gamma_0$

First difference: $\Delta y_t = y_t - y_{(t-1)}$

Backshift: $B y_t = y_{(t-1)}$

PYQ Questions From Unit 3 With Answers

Q. PYQ: Explain white noise and its conditions. (5 marks)

White noise is a purely random time series process with no systematic pattern.

Its mean is zero or constant, variance is constant, and autocorrelation is zero at all non-zero lags.

In time series modelling, residuals should behave like white noise.

If residuals are white noise, it means the model has captured most of the useful pattern.

Exam line: Write definition + 3 conditions + residual importance.

Q. PYQ: Explain ACF and PACF and their use in model identification. (5 marks)

ACF measures correlation between values of a time series separated by different lags.

PACF measures direct correlation at a lag after removing the effect of shorter lags.

For AR(p), PACF cuts off after lag p and ACF tails off.

For MA(q), ACF cuts off after lag q and PACF tails off.

For ARMA, both ACF and PACF usually tail off.

Exam line: End with: ACF and PACF help choose p and q before fitting ARMA/ARIMA.

Q. PYQ: Prove first-order differencing can make $y_t = \beta_0 + \beta_1 t + e_t$ stationary. (5 marks)

Given: $y_t = \beta_0 + \beta_1 t + e_t$

Previous value: $y_{(t-1)} = \beta_0 + \beta_1(t-1) + e_{(t-1)}$

Difference: $y_t - y_{(t-1)} = \beta_1 + e_t - e_{(t-1)}$

The time trend t is removed after differencing.

Therefore first-order differencing can remove linear trend and help make the series stationary.

Exam line: This is a direct compulsory-style proof.

Q. PYQ: Differentiate autoregression and moving average models. (5 marks)

Autoregression: uses past values of the same series.

Moving Average: uses past error terms or shocks.

AR example: $y_t = c + \phi_1 y_{(t-1)} + e_t$

MA example: $y_t = \mu + e_t + \theta_1 e_{(t-1)}$

AR is about old actual values; MA is about old mistakes/errors.

Exam line: This was asked as AR vs MA in the current paper.

Solved Numerical: Autocovariance And Autocorrelation

PYQ type: Given a time series, compute mean, variance, autocovariance at lag 1, 2, 3, and autocorrelation at lag 1, 2, 3.

Numerical Method

Step 1: Find mean $\bar{x}$.

Step 2: Find variance $\gamma_0 = \sum[(x_t - \bar{x})^2] / n$.

Step 3: For each lag k , find $\gamma_k = \sum[(x_t - \bar{x})(x_{t-k} - \bar{x})] / n$.

Step 4: Find $r_k = \gamma_k / \gamma_0$.

Step 5: If r_k is positive, series shows positive correlation at that lag.

Lag	Autocovariance	Autocorrelation	Comment
1	7.766	0.788	Strong positive
2	4.677	0.475	Moderate positive
3	2.505	0.254	Weak positive

For current paper data $X = [9, 7, 8, 10, 12, 11, 13, 15, 14, 16, 17, 15]$, mean = 12.25 and variance = 9.854 using denominator n . Since all three autocorrelations are positive, the series shows positive correlation at lags 1, 2 and 3.

Last-Day Unit 3 Revision

- White noise: random series, constant mean, constant variance, zero autocorrelation.
- Stationarity: constant mean, constant variance, autocovariance depends only on lag.
- Differencing: removes trend; first difference is $y_t - y_{t-1}$.
- Backshift: $B y_t = y_{t-1}$.
- AR: uses past values. MA: uses past errors.
- ACF/PACF: PACF cut means AR; ACF cut means MA.
- Autocorrelation numerical: mean $\rightarrow$ variance $\rightarrow$ autocovariance $\rightarrow$ autocorrelation $\rightarrow$ comment.

End of Unit 3 Notes